SAMPLE

Cosmic Explorers: Space Telescopes

Hubble, Webb, and What They Saw

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Introduction

A telescope on the ground is a compromised instrument. It looks through a swirling ocean of air that blurs, blocks, and scatters the light arriving from space; it can observe only at night, only when clouds permit, and only in the narrow spectral windows the atmosphere happens to leave open. The space telescope, placed above all of this, is astronomy's purest eye, and this book is its account: why we put telescopes in space, how we learned to do it, what the great instruments have found, and what the coming decades of space observatories promise.

The idea predates the space age, but only the rocket could realize it. Within a generation of the first satellites, astronomers had placed ultraviolet and X-ray detectors above the atmosphere and discovered a universe invisible from the ground; within two, the Hubble Space Telescope had delivered the sharpest views of the cosmos ever seen and transformed the telescope from a tool of observation into an instrument of public wonder; and in the present generation, the James Webb Space Telescope, the largest telescope ever flown, is reading the infrared light of the first galaxies, the atmospheres of other worlds, and the interiors of star-forming clouds.

The book's plan follows the subject's logic. The opening chapters give the foundations: the atmosphere's limits that motivate the enterprise, the pioneers who imagined observatories in orbit, and the engineering that makes them possible. The central chapters give the instruments and their discoveries: Hubble's story, the infrared observatories that see the cold and the distant, the high-energy telescopes that see the violent universe, the planet hunters that census other worlds, and the Webb, the current flagship. The closing chapters widen the view: the engineering and operations of the observatories themselves, the future missions on the drawing boards, and the space telescope's place in the human adventure of knowing the universe.

A theme runs throughout: the space telescope is the meeting point of astronomy and engineering, and its history is a history of both. Every instrument in this book required a rocket, a mirror, a detector, a control system, and a repair or a workaround; every discovery was enabled by an engineering triumph, and several missions, Hubble above all, were saved by engineering heroics after engineering failures nearly ended them. Anyone who finishes this book will understand the space telescope as the most complete expression of the scientific instrument maker's art, a machine that carries the oldest human curiosity, the wish to see, to the very edge of what can be seen.

This subject opens with a fact worth fixing first, one that explains why the entire discipline moved into space. The atmosphere is an astronomer's worst colleague: it blurs the light, devours the ultraviolet and the infrared, and blocks the X-rays and the gamma rays entirely, so the observatories on the ground see only a narrow window of the universe's full message, and the space telescope, in the plain accounting, is a machine for reopening the book. The reader is opening a history of the instruments that took astronomy out of the atmosphere, and the chapters that follow move chronologically through the machines that remade the science, from the first sounding rockets to the great observatories of the coming decades.

A second framing observation sets the book's tone, and it is the one every mission chapter will repeat. A space telescope costs, roughly, a decade of national effort and a budget measured in the billions, it flies on engineering margins of millimeters and kelvin, and it can never be serviced unless the design says otherwise, which makes the discipline a high-stakes cousin of astronomy proper, part optics and part expedition; the reader should look forward, across the chapters, to stories of engineering courage at least equal to the science, because in this subject the engineering is what makes the science possible, and the failures, when they come, are expensive enough to make the history read at times like an adventure.

Chapter 01: The Atmosphere's Limits: Why Space

The space telescope begins with a problem, and the problem is the air. This chapter describes what the atmosphere does to astronomical light, why the ground-based telescope is fundamentally compromised, and what the observatory above the atmosphere gains.

The atmosphere's first offense is absorption. The air is opaque to most of the electromagnetic spectrum: the atmosphere's gases absorb the ultraviolet below about 300 nanometers, blocking the ozone layer's window; the X-rays and the gamma rays, absorbed in the upper atmosphere, never reach the ground at all; and the infrared, absorbed by the water vapor and the carbon dioxide, is observable only in a few narrow bands between the molecular absorption lines. The visible window, the band from about 300 to 700 nanometers that the eye shares, and the radio bands, are the atmosphere's only generous openings, and the ground-based astronomer works almost entirely within them. The refraction's second face deserves a paragraph, since the air moves the light as

well as blurs it. The atmosphere's bending, the refraction that lifts the stars near the horizon, shifts the apparent positions by the arcsecond's amounts, and the shift varies with the temperature and the pressure, the positional astronomy's noise; the dispersion, the wavelength's dependence of the bending, spreads the star's light at the low elevations, the blue and the red bent differently; and the photometric variability, the cloud's unseen absorption and the aerosols' scintillation, limits the brightness measurements' precision. The reader should note the astrometry's stake: the micro-arcsecond measurements of the Gaia era demanded the space environment, the positions' wobble-free measurement, and the atmosphere's refraction, the positional astronomy's oldest correction, is simply absent above the air.

The reader should picture the situation from the cosmos's side: a universe glowing at every wavelength, and a planet's air that lets through two thin slices of the whole.

The atmosphere's second offense is blurring. The air is turbulent: the pockets of differing temperature and density, carried by the winds through the telescope's line of sight, refract the light in shifting amounts, and the image, gathered over seconds and minutes, is smeared. The seeing, the astronomer's term for the blur, limits the ground-based telescope's resolution to roughly one arcsecond regardless of the mirror's size: a 10-meter telescope on a mediocre night resolves no better than a 20-centimeter telescope, the atmosphere imposing its ceiling on every instrument beneath it. The light pollution deserves its paragraph, since the ground's brightening is the space's case's newest argument. The cities' light, the sodium's and the LED's glow scattered by the atmosphere, has brightened the night sky over the observatories' sites: the mountain peaks of the twentieth century's darkness now sit under the domes' haze, the wilderness of Chile and Hawaii protected by the lighting ordinances the astronomers fought for; and the growth's trajectory, the population's and the development's expansion, makes the ground's darkness a managed and diminishing resource. The reader should note the space telescope's freedom: the orbit's night, the permanent and the unchanging, is the darkness the ground can only approximate, and the faint-object science, the deep fields and the surface brightness work, is increasingly the orbit's monopoly, the atmosphere's transparency not the only sky condition that matters.

The adaptive optics of the modern era, the systems that measure and partially cancel the turbulence, have narrowed the gap, but the space telescope's freedom from the air remains complete, and the reader should hold The diffraction limit's arithmetic deserves a paragraph, since the sharpness is the physics's gift. The telescope's resolution, the finest angle its aperture can distinguish, is set by the light's wavelength divided by the mirror's diameter: the bigger the mirror at a given wavelength, the finer the view, and the Hubble's 2.4 meters at the visible light deliver the twentieth of an arcsecond, the ground's seeing exceeded tenfold; the same physics in the infrared, the longer wavelengths, demands the bigger mirrors, the Webb's 6.5 meters at the two micrometers matching the Hubble's sharpness at its own bands, the aperture's growth the wavelength's compensation. The reader should hold the number's meaning: the space telescope's sharpness is not the atmosphere's absence alone but the physics's optimum, the aperture's full value unlocked by the orbit's stability, and every mission of this book is, at bottom, an argument about the wavelength and the diameter.

the comparison's essential numbers: Hubble, a 2.4-meter telescope, resolves about ten times finer than the best ground-based seeing of its era, and the gain comes entirely from the absence of the air.

The third offense is brightness, the sky's own. The air glows: the scattered sunlight, the moonlight, the artificial lights of the cities bounced off the atmosphere's aerosols, and the air's own emission, the excited atoms' natural lines, raise the night sky's brightness and set a floor beneath which the faint objects disappear. The space telescope, above the airglow and on the night side of its orbit, works against a sky hundreds of times darker, and the difference decides what can be seen: the deepest Hubble exposures, staring at a single patch of sky for days of accumulated time, reach objects a hundred times fainter than the sky from the ground, and the faintest galaxies, the earliest and the most distant, are visible only from the darkness above the atmosphere.

A note on the wavelength census belongs here, since the chapter's limits define the whole book's map. The ground's full access: the visible and the radio, the two windows the atmosphere grants; the ground's partial access: the infrared's and the submillimeter's bands, the dry sites' windows; the space's exclusive domain: the ultraviolet, the X-rays, the gamma rays, the entire far infrared, and the precision astrometry and photometry the air's unsteadiness forbids. The reader should hold the map's meaning: the space telescope is not merely the ground telescope's superior but the ground telescope's complement, the wavelengths and the precisions the ground cannot reach, and the book's chapters, organized by the wavelength and the mission, are the map's territories, each one the atmosphere's surrender.

The chapter's summary is the space telescope's charter. Above the atmosphere, the observatory gains the whole

spectrum, the diffraction-limited sharpness of its optics, and a sky nearly as dark as space itself; the costs are the launch's expense, the instrument's inaccessibility, and the unforgiving environment of space. The rest of the book is the story of paying those costs, and of what the purified eye has seen.

The limits chapter closes on the ground-based counterpoint, because the book's case for space needs its honest frame. The atmosphere that blocks the spectrum also steadies the image, the mountaintops that lose the infrared still hold the biggest mirrors, and the modern techniques, adaptive optics, lucky imaging, have kept the ground competitive in the visible and even winning in aperture; the reader should file the subject as a partnership, with space opening the windows the ground cannot reach and the ground supplying the light-gathering power space cannot afford, and the astronomy of the current era, in every wavelength chapter that follows, is a choreography between the two.

The numbers make the case vividly. A modest orbiting instrument delivers images three to five times sharper than the finest ground site on the finest night, and it delivers that sharpness on every target, every night, for years, while the ground campaign waits for the one night in a season when the air stills. Photometry from orbit reaches precisions the ground cannot match even in aggregate, which is why the transiting-planet science of recent decades moved wholesale above the air.

Absorption bars whole disciplines from the ground regardless of seeing. Water vapor and carbon dioxide swallow the infrared above a wavelength of about one micrometer, ozone blocks the ultraviolet below 320 nanometers, and the atmosphere is entirely opaque to X-rays and gamma rays, so the cold universe, the hot universe, and the young universe are simply not visible from any mountaintop. The space observatory is not an upgrade to ground astronomy; it is access to light that no ground instrument has ever physically received.